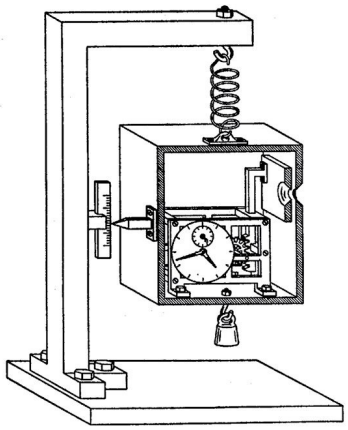


Figure 4.8 The ‘light box’ in Einstein’s mental experiment, as drawn by Bohr.

This is the vertiginous dive taken by Bohr, Heisenberg and Dirac – into the depth of the nature of things.

But do we really understand?

Certainly, quantum mechanics is a triumph of efficacy. And yet … are you sure, dear reader, that you have fully understood what quantum mechanics reveals to us? An electron is nowhere when it is not interacting … mmm … things only exist by jumping from one interaction to another … well … Does it all seem a little absurd?

It seemed absurd to Einstein.

On the one hand, Einstein proposed Werner Heisenberg and Paul Dirac for the Nobel Prize, recognizing that they had understood something fundamental about the world. On the other, he took every opportunity to grumble that, however, none of this made much sense

The young lions of the Copenhagen group were dismayed: how could this come from *Einstein* himself? Their spiritual father, the man who had the courage to think the unthinkable, now pulled back and feared this new leap into the unknown – the very leap which he had himself triggered. How could it be that the same Einstein, who had taught us that time is not universal and that space bends, was now saying that the world could not be *this* strange?

Niels Bohr patiently explained the new ideas to Einstein. Einstein objected. Bohr, in the end, always managed to find answers to the objections. The dialogue continued for years, by way of lectures, letters, articles … Einstein devised mental experiments to show that the new ideas were contradictory: ‘Imagine a box filled with light, from which is let escape for a brief instant a single photon …’: thus one of the most famous examples of these begins (figure 4.8).[fn28]

During the course of the exchange, both great men had to give way, to alter their ideas. Einstein was obliged to recognize that there was actually no